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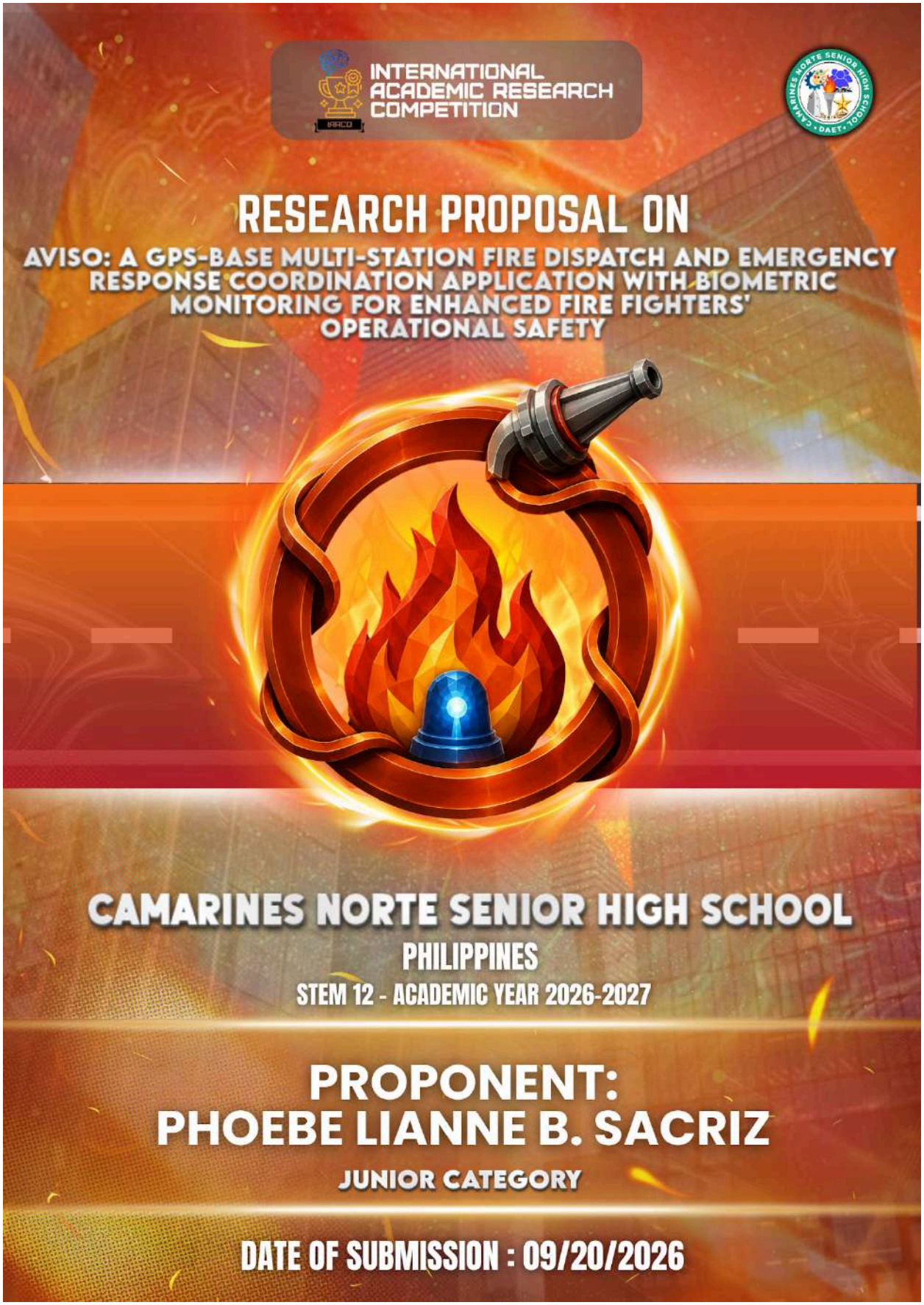
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INTERNATIONAL ACADEMIC RESEARCH COMPETITION
IARCO
CAMARINES NORTE SENIOR HIGH SCHOOL • DAET •
RESEARCH PROPOSAL ON
AVISO: A GPS-BASE MULTI-STATION FIRE DISPATCH AND EMERGENCY RESPONSE COORDINATION APPLICATION WITH BIOMETRIC MONITORING FOR ENHANCED FIRE FIGHTERS' OPERATIONAL SAFETY
CAMARINES NORTE SENIOR HIGH SCHOOL
PHILIPPINES
STEM 12 - ACADEMIC YEAR 2026-2027
PROPONENT:
PHOEBE LIANNE B. SACRIZ
JUNIOR CATEGORY
DATE OF SUBMISSION : 09/20/2026

IARCO RESEARCH PROPOSAL

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Research Topic: AVISO: A GPS-Based Multi-Station Fire Dispatch and Emergency Response Coordination Application with Biometric Monitoring for Enhanced Firefighters' Operational Safety

Research Problem

Fire emergencies continue to be one of the most critical public safety issues in the world today, taking lives, destroying the environment and costing billions of dollars of property damage each year. Yet, despite these concerns the Philippines Bureau of Fire Protection (BFP) still lacks modern technology in terms of communication, health monitoring of firefighters such as pulse rate, and real time GPS monitoring. Furthermore, they still rely on handheld transceivers for communication instead of using advanced communication, which can lead to miscommunication and misunderstanding during fire emergencies that require more enhanced communication, navigation, and information technologies to improve coordination and operational effectiveness during fire emergencies [1].

22 To address these gaps, this study will develop AVISO: A web-based application that integrates a GPS location tracking, multi-station alert, and faster route search for enhanced firefighters' emergency dispatch, response and coordination. Furthermore, AVISO uses a one-tap back-up call system and Biometric monitoring for heightened BFP personnel safety during fire emergencies.

Existing Literature

The Bureau of Fire Protection (BFP) currently faces operational bottlenecks due to a reliance on handheld transceivers, a lack of real-time GPS tracking for responding units, and an absence of vital sign health monitoring for deployed personnel [1].

GPS/GIS-based routing systems demonstrate that real-time traffic and location data can shorten response times, though adoption outside pilot settings remains limited [2][3].

Wearable and IoT-based monitoring technologies show that continuous tracking of firefighter vitals and environmental hazards is both feasible and effective for protecting responders in the field, while broader adoption studies confirm growing interest tempered by cost, organizational readiness, and user acceptance barriers [4][5][6][7][8].

Fire station planning and capability evaluation models address the upstream question of how response infrastructure itself should be sited and assessed, ensuring resources are positioned to match actual risk rather than administrative convenience [9][10]. Communication and coordination research consistently points to the same recurring problem across contexts: delayed or unclear information flow between responders, dispatchers, and the public, which structured protocols and direct-to-nearest-unit notification help to reduce [11][12][13].

Digital twin applications extend this further by simulating real-time conditions for facility management and occupant evacuation, though data integration, cost, and low technical literacy remain unresolved barriers to adoption [14][15]. At the local level, community-based systems

show that even simple GPS- and SMS-based tools can close the coordination gap between residents and barangay-level responders, achieving high user acceptance in Philippine settings [16][17]. At the same time, prior work highlights the real-world challenges any new system must address — data integration across health and environmental sources, communication reliability in complex or high-rise terrain, and organizational readiness for adoption — which inform this study's design and evaluation priorities [2][7][12][14].

Research Question

This study aims to develop a mobile application that improves fire emergency dispatch coordination and firefighter safety monitoring during rescue operations. Specifically, this study seeks answers to the following questions:

1. How effective is AVISO in terms of:

a. Multi-Station Alert Broadcasting b. GPS Location Tracking of Fire Units c. Response Time d. Biometric (Pulse Rate) Monitoring e. One-Tap Backup Calling f. Notification Accuracy to Admin/Ground Commander

2. Is there a significant difference in the time of sending alert signals and backup request by AVISO across the BFP stations through various trials conducted in:

a. Daet b. Basud c. Mercedes

3. What is the rate of acceptability of the developed web-mobile application among BFP personnel, MDRRMO responders, and Red Cross representatives based on the ISO/IEC 25010 software quality standards in terms of:

a. Functional Suitability b. Performance Efficiency. c. Usability d. Reliability e. Security

Methodology

The study will employ a quantitative developmental and experimental approach to develop, test, and evaluate AVISO.

1. **System Development:** AVISO will be developed as a web and mobile application using React.js, React Native, Laravel, and MySQL. It will provide multi-station alert broadcasting, GPS tracking of fire units, response-time monitoring, smartwatch-based pulse monitoring, one-tap backup requests, and real-time notifications. A cloud/VPS server will host the backend and database, while Axios will connect the applications to the system.

2. **System Integration:** The researchers will integrate the GPS, notification, backup request, and smartwatch pulse-monitoring functions into AVISO. The system will be configured for

authorized BFP personnel and undergo debugging and initial API testing using Postman before actual experimentation.

3. **Testing and Evaluation:** AVISO will undergo initial system testing and controlled mock-fire testing. Initial system testing will assess alert transmission, GPS tracking, pulse-data transmission, backup requests, and notification delivery. Field testing will involve three mock-fire trials on separate days, using one BFP firetruck per trial. The researchers will record alert transmission time, GPS accuracy, response time, biometric accuracy, backup request transmission, and notification accuracy.

4. **Acceptability Rating:** Selected BFP personnel, MDRRMO responders, Red Cross representatives, and a system. The developer will evaluate AVISO using an ISO/IEC 25010-based four-point rubric covering Functional Suitability, Performance Efficiency, Usability, Reliability, and Security.

5. **Data Analysis and Calibration:** The collected data will be analyzed using mean, standard deviation, weighted mean, and error analysis. ANOVA will be applied when appropriate for comparing multiple independent testing groups or conditions. Results will be used to calibrate the system and formulate recommendations for improving AVISO's reliability, functionality, and operational effectiveness.

Expected Outcome

The study is expected to demonstrate the effectiveness of AVISO as a GPS-based multi-station fire dispatch and emergency response coordination application with biometric monitoring. It will evaluate the system's alert transmission, GPS tracking, response time, pulse monitoring, backup requests, and notification accuracy. The results will guide improvements in firefighter communication, coordination, and operational safety.

Potential Limitation

The application may experience Bluetooth interference and intermittent disconnection between the smart watch and mobile application in Daet, Basud, and Mercedes due to varying signal strength and terrain interference in these areas. Additionally, the biometric monitoring feature is constrained by the limited range of Bluetooth connectivity, requiring the firefighter's device to remain within close proximity to the smart watch, as well as the limited battery life of the smart watch, which may not sustain continuous pulse rate monitoring throughout prolonged fire operations.

Conclusion

AVISO responds directly to the gaps identified by BFP Daet personnel themselves — the continued reliance on handheld transceivers, the absence of real-time GPS visibility over responding units, and the lack of any form of health monitoring for firefighters in the field. By combining multi-station alert broadcasting, GPS-based dispatch and route tracking, smartwatch-based biometric monitoring, and a one-tap backup-calling mechanism within a single web-and-mobile platform, the system is designed to close these gaps rather than address them in isolation.

Grounded in the Technology Acceptance Model, Emergency Management Theory, and Task-Technology Fit Theory, AVISO is expected to be adopted readily by its intended users because its features are built directly around operational needs firefighters and dispatchers already described: faster alerting, clearer coordination across the Daet, Basud, and Mercedes stations, and a simple, low-effort way to call for backup or flag a medical distress event during an active response.

Through controlled mock-fire trials and evaluation against the ISO/IEC 25010 software quality standards, the study intends to generate concrete, measurable evidence — alert transmission speed, GPS tracking accuracy, response time, biometric accuracy, and notification accuracy — of whether the system delivers on this goal. In doing so, AVISO supports SDG 11 (Sustainable Cities and Communities) by strengthening the responsiveness and resilience of local emergency services, and offers the Bureau of Fire Protection a low-cost, locally grounded model for modernizing fire dispatch and firefighter safety monitoring in similarly under-resourced municipalities.

Project Practicalities

Development team and partners. The study is carried out in direct partnership with the Bureau of Fire Protection (BFP) Daet Fire Station, which supplies the operational context, personnel, and fire truck units used in testing. A system developer is consulted from the outset to validate the technical feasibility of the proposed architecture, and BFP personnel are interviewed early on to ground the application's feature set in real dispatch and coordination problems rather than assumed ones.

Technology stack and infrastructure. AVISO is built as a web application (React.js) and a companion mobile application (React Native), backed by a Laravel framework and a MySQL database, communicating over RESTful APIs via Axios. A cloud/VPS server hosts the backend and database for real-time data exchange. Development relies on a workstation with at least an Intel Core i5 or Ryzen 5 processor, 16 GB RAM, and a 256 GB SSD, alongside Node.js, npm, Composer, Git/GitHub for version control, Postman for API testing, and Tailwind CSS for the interface.

Testing constraints. Because BFP Daet has only two fire truck units, field testing is deliberately limited to three separate mock-fire trials conducted on different days at Camarines Norte Senior High School (CNSHS), each using a single fire truck and crew, so that live emergency operations are never disrupted. For the same safety reason, no data is collected during actual fire emergencies; all trials are controlled simulations.

Scope constraints. The mobile application is developed exclusively for Android devices (version 5.0 and above); iOS is not supported within this study. AVISO also requires an active internet connection, GPS, and Bluetooth to function, and access is restricted to verified BFP Administrator and Ground Commander accounts to protect operational plans and firefighter health data. The system is scoped to major fire emergencies only (house, building, and grass fires) and is not designed to replace BFP's official firefighting techniques or chain of command.

Timeline. The research and development activities — from interview and consultation through layout and design, materials gathering, coding, integration, testing, calibration, final testing, and acceptability evaluation — are planned to be carried out within School Year 2026–2027.

Post-Program Plan

Calibration and hand-off. Following final testing, identified issues (such as GPS tracking inaccuracies, notification delays, or unstable smartwatch-to-application transmission) are corrected through an iterative calibration cycle, with the goal of leaving BFP Daet, Basud, and Mercedes with a stable, field-ready version of AVISO rather than a one-time prototype.

Scaling within the province. The pilot is deliberately structured across three municipalities; Daet, Basud, and Mercedes — specifically to generate evidence on how the system performs across varying distances and station conditions. A successful pilot is intended to support a case for extending AVISO to the remaining Bureau of Fire Protection stations across Camarines Norte, using the same multi-station alert and GPS dispatch architecture already built and tested.

Addressing known limitations. Post-pilot development work will prioritize the constraints already identified during the study, Bluetooth range and connection stability between the smartwatch and the mobile application, and the smartwatch's limited battery life during prolonged operations with options such as extended-range pairing, low-power monitoring modes, or alternative wearable hardware to be explored in a future iteration.

Institutional adoption. Because access to AVISO is restricted to verified BFP Administrator and Ground Commander accounts, the system is positioned to be formally adopted into the station's existing dispatch workflow rather than used as a parallel or experimental tool, supporting the recommendation, echoed in related local studies that community and inter-agency coordination with the Municipal Disaster Risk Reduction and Management Office (MDRRMO) and the Philippine Red Cross be formalized alongside the technology itself.

Direction for future research. Consistent with the study's own significance section, the researchers intend for AVISO's design and evaluation results to serve as a reference for future researchers extending the platform beyond fire incidents (e.g., to other emergency types) or beyond Camarines Norte, and for iOS support to be considered once the Android-based pilot has demonstrated consistent reliability across the three test municipalities.

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